INCH-POUND
MIL-DTL-7793/14B
22 May 2017
SUPERSEDING
MIL-DTL-7793/14A
19 October 2012

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL,
28 VDC OR 26 VAC, 50 TO 2400 Hz, SQUARE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

Reactivated after 19 October 2012 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-7793.

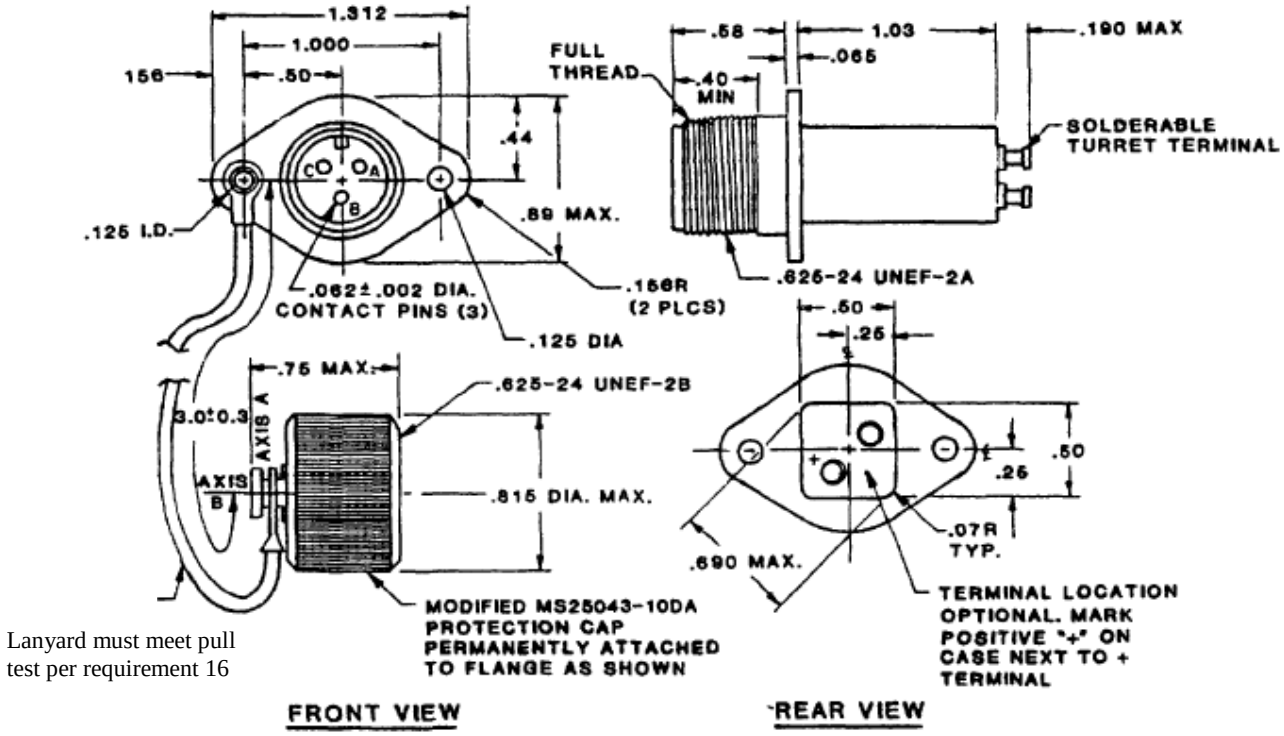


FIGURE 1. Meter dimensions.

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REQUIREMENTS:

1. Dimensions. Dimensions are as specified on figure 1. All dimensions are in inches. Unless otherwise specified, tolerances are ±0.02 for two-place decimals and ±0.015 for three-place decimals.

2. Mounting flange. The mounting flange, threaded extension and threaded protection cap shall be constructed of aluminum alloy with an olive drab chromate coating over cadmium plating.

3. Protection cap. The protection cap shall be in accordance with MS25043-10DA per SAE AS25043 except for the finish and means of attachment. The finish shall be in accordance with that for the mounting flange in requirement 2. The metallic chain shall be replaced with a nylon lanyard.

4. Dielectric strength. The meter shall be capable of withstanding the application of 600 VRMS (room) and 350 VRMS (altitude) between the terminals and the case.

5. Life accuracy. Meter accuracy shall be ±0.1 percent from -85 to 257° F (-65 to 125° C) and 10 to 34 VDC, or 20 to 30 VAC at 50 to 2400 Hz. Qualification tests shall be performed on 3 meters each, for DC and AC accuracy.

5. Power consumption. Power consumption at 28 volts VDC shall not exceed 50 milliwatts and at 26 VAC, 400 Hz AC shall not exceed 25 milliwatts.

6. Ripple voltage. The meter shall continue to operate as specified in MIL-DTL-7793 when subjected to a cyclic peak of ripple voltage (see Note 3) of less than 2.0 VDC and the frequency-voltage coordinates on figure 2.

6.1 Ripple voltage measurement. The ripple voltage shall be measured with a peak reading voltmeter in series with a 4.0-microfarad capacitor. The higher of the two values measured when the voltmeter is successively connected to each of two polarities shall be considered the ripple voltage.

6.2 Frequency characteristics. The frequency components of the ripple voltage shall be within the limits on figure 2 when measured as conducted interference.

7. Operating temperature range. The operating temperature range shall be from -85 to 257° F (-65 to 125° C).

8. Part Number. The part number is M7793/14-1.

9. Time range. Total readout of the meters shall be 99999.99 hours.

10. Meter output. The output from the meter shall be in a serial binary coded decimal format. The output impedance shall be 100K ±10 percent. A logical “zero” shall be between 0.0

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to +0.2 volt. A logical “one” shall be between +3.3 and +6.6 volts. The output pin assignments are shown in Figure 3 and the output data format as shown on figure 4.

11. Transient protection. No temporary or permanent degradation shall be produced in the meter for the input voltage and time values shown on figure 5 and figure 6 (80V and 600V transient, respectively).

11.1 Transient voltage. Transient surge voltages shall be within the limits on figure 3 when converted to their equivalent step functions. Transient spike voltages shall be within the limits shown on figure 6. They shall be not greater than ±600V nor depart from the normal wave form for more than 50 microseconds.

12. Durability of cap and threads. There shall be no mechanical or electrical degradation after 100 cycles in accordance with SAE AS50151.

13. Voltage range. The operating voltage shall be from 10 to 34 VDC or 20 to 30 VAC, 50 to 2400 Hz.

14. Durability of connectors and contacts. No mechanical or electrical degradation shall occur to connectors or contacts after 100 cycles in accordance with SAE AS50151. Operational accuracy shall be checked at the end of the test.

15. Weight. The weight shall be not greater than 1.0 ounce with the attached protection cap.

16. Tensile strength of lanyard. The lanyard shall withstand a tensile load of 25 pounds when applied to Axis “A” and Axis “B” as shown on figure 1.

17. Environmental thermal stress screening test profile.
Temperature range: -65° C to +125° C
Temperature rate of change: 2° C/minute minimum (chamber air temperature)
Temperature dwell duration: 30 minutes minimum
Temperature cycles: 5 cycles minimum (one cycle consists of one dwell at each temperature extreme)
Power on/Equipment operating: Power meter at 10 VDC or 26 VAC 400 Hz. The meter shall be turned on and off a minimum of three times at each temperature extreme (-65° C and +125° C) for each cycle.

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18. Qualification tests. Qualification tests shall be conducted in accordance with MIL-DTL-7793, except that the accuracy, voltage, temperature, and thermal shock or environmental stress screening shall be as specified herein. Unless otherwise specified, all qualification tests, which will require the units to be powered, may be performed using either AC or DC power. In lieu of Thermal Shock, 100 percent Environmental Thermal Stress Screening may be performed on 100 percent of meters supplied under this specification sheet. Salt spray, moisture resistance, vibration, shock, electromagnetic compatibility and low temperature storage shall be conducted with the protective cap in place. Hermetic seal test is not applicable.

19. Conformance inspection. Conformance inspections shall be conducted in accordance with MIL-DTL-7793, except that the voltage, temperature, and thermal shock or environmental stress screening shall be as specified herein. In lieu of thermal shock, 100 percent environmental thermal stress screening may be performed on 100 percent of meters supplied under this specification sheet. Hermetic seal test is not applicable. Unless otherwise specified, all conformance tests require the units to be powered using either AC or DC power.

NOTES:

1. Elapsed time is read from the M7793/14-1 meter by mating the M7793/14-1 meter's connector with the M7793/12-1 reader's connector and operating the reader.
2. Digital readout and meter face plate per MIL-DTL-7793 are not applicable since the digital readout is provided by the separate M7793/12-1 reader.
3. AC operation may be from either sinusoidal or square wave sources.
4. The meter is designed to have an electrical ground and a continuous EMI/RFI shield with the treaded protection cap in place.
5. Contacts shall meet all the requirements of silver plated solder contacts from SAE AS50151.

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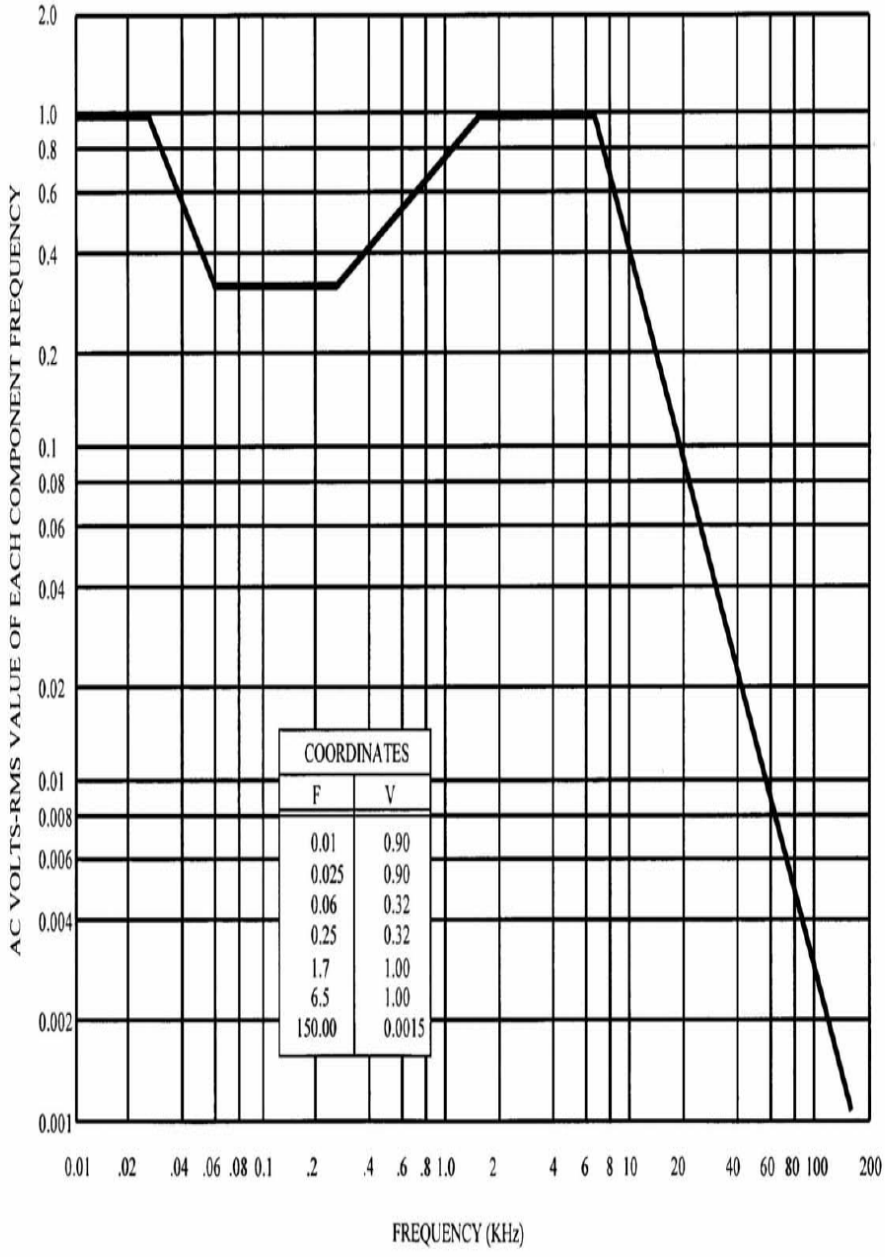


FIGURE 2. Frequency characteristics of ripple in 28 VDC electric systems.

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FIGURE 3. Readout pin assignment.

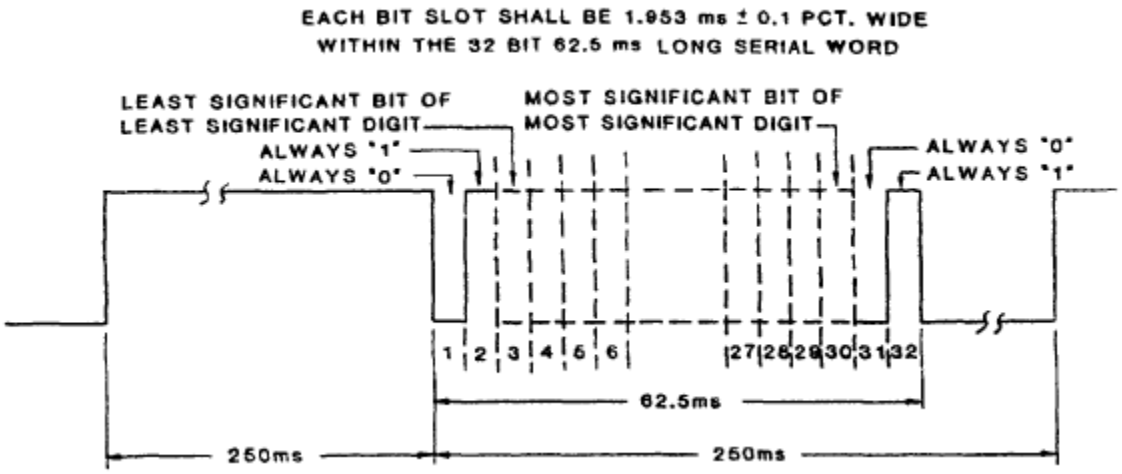


FIGURE 4. Output data format.

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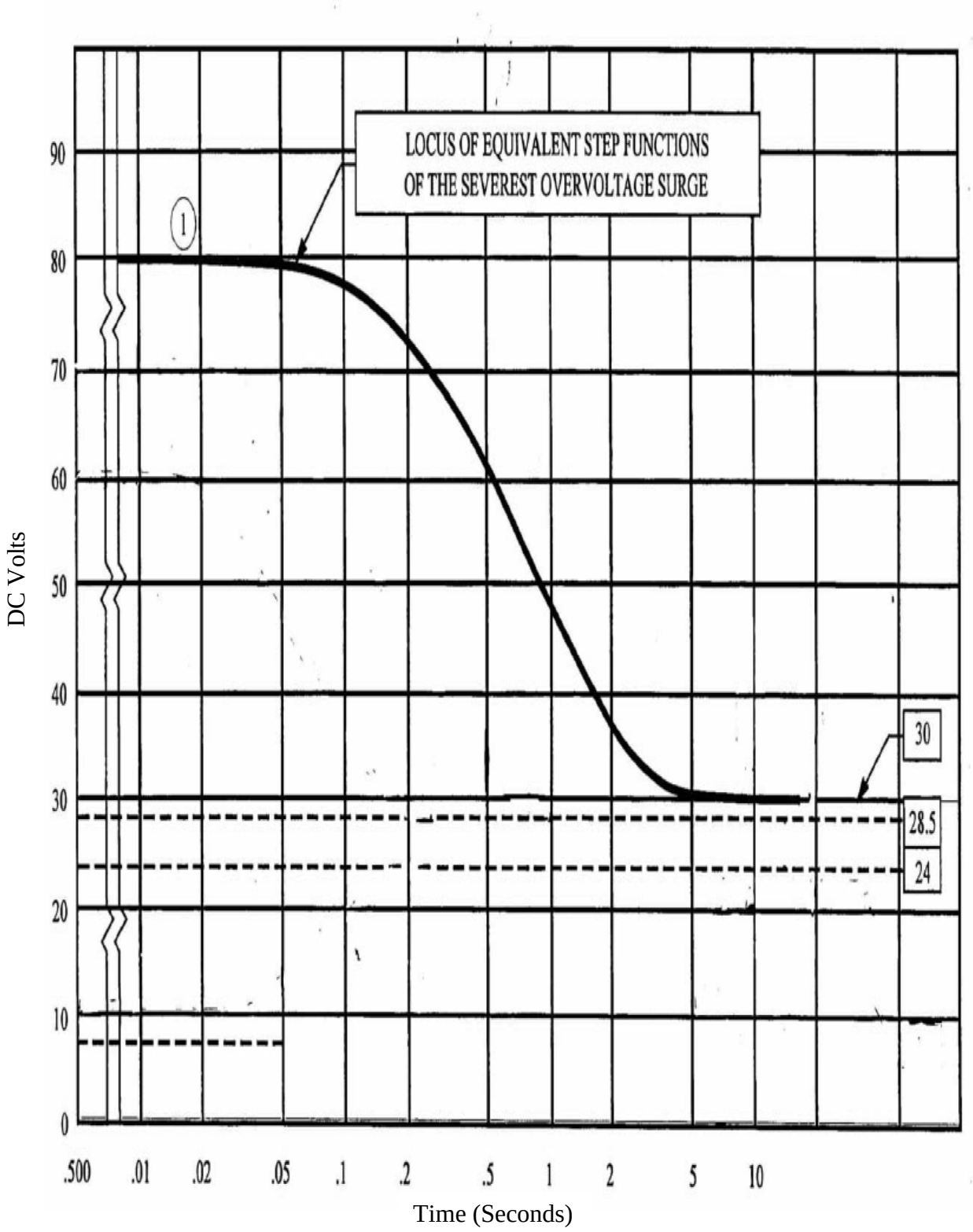


FIGURE 5. Transient surge step function loci limits.

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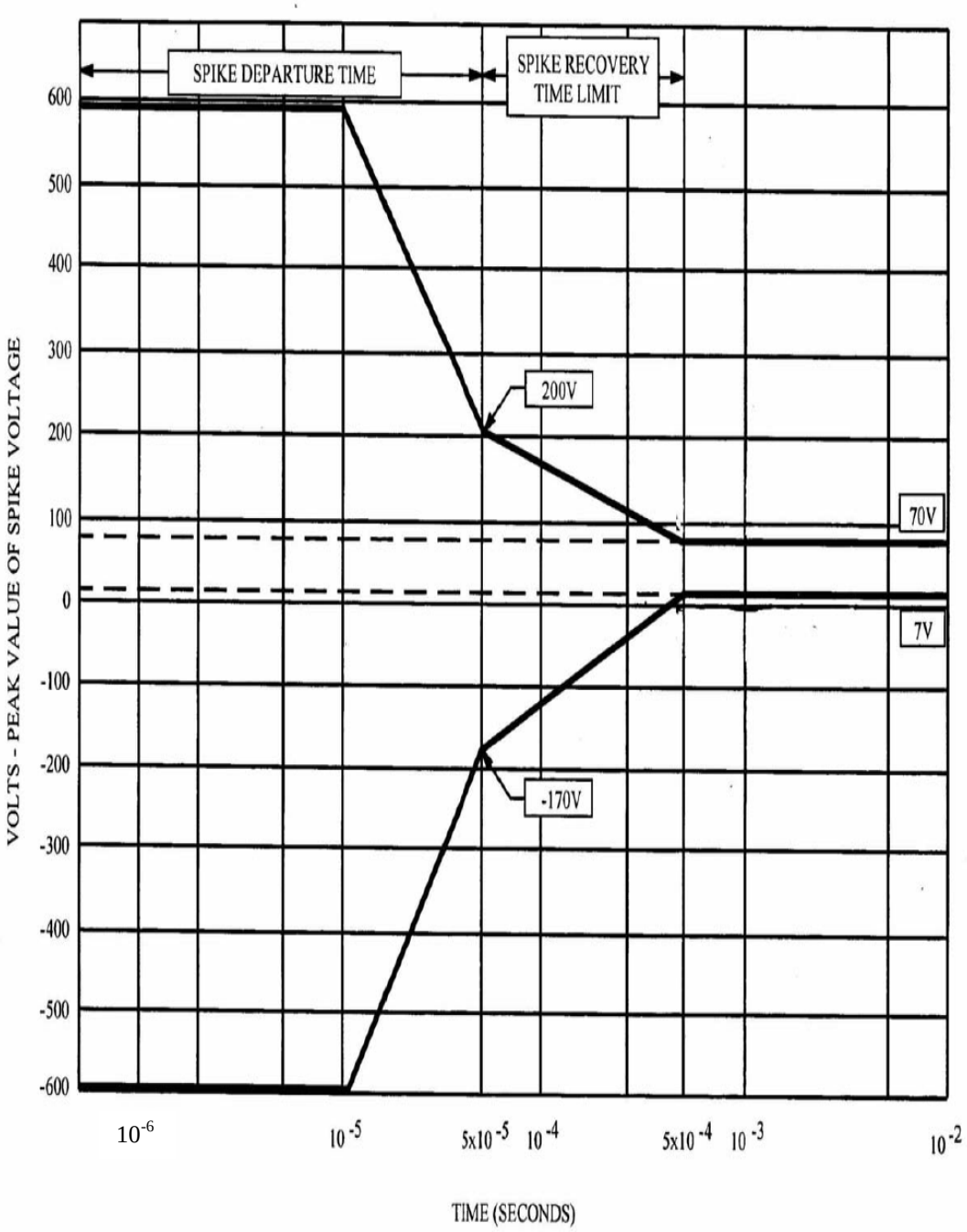


FIGURE 6. Envelope of spike voltages for direct current equipment.

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Referenced documents. In addition to MIL-DTL-7793, this specification sheet references SAE AS25043 and SAE AS50151.

Custodians:
Army - MI
Navy - AS
Air Force - 71

Preparing activity:
Navy - AS
(Project 6645-2017-002)

Review activities:
Navy - OS, SH
DLA - GS 1

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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